

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

Question

|     |        |
|-----|--------|
| $x$ | $f(x)$ |
| 1   | $a$    |
| 2   | $a^5$  |
| 3   | $a^9$  |

For the exponential function  $f$ , the table above shows several values of  $x$  and their corresponding values of  $f(x)$ , where  $a$  is a constant greater than 1. If  $k$  is a constant and  $f(k) = a^{29}$ , what is the value of  $k$  ?

Rationale

The correct answer is 8. The values of  $f(x)$  for the exponential function  $f$  shown in the table increase by a factor of  $a^4$  for each increase of 1 in  $x$ . This relationship can be represented by the equation  $f(x) = a^{4x+b}$ , where  $b$  is a constant. It's given that when  $x = 2$ ,  $f(x) = a^5$ . Substituting 2 for  $x$  and  $a^5$  for  $f(x)$  into  $f(x) = a^{4x+b}$  yields  $a^5 = a^{4(2)+b}$ . Since  $4(2) + b = 5$ , it follows that  $b = -3$ . Thus, an equation that defines the function  $f$  is  $f(x) = a^{4x-3}$ . It follows that the value of  $k$  such that  $f(k) = a^{29}$  can be found by solving the equation  $4k - 3 = 29$ , which yields  $k = 8$ .